

CAMPINA GRANDE, Brazil

WMO: 827950

Lat: 7.270S

Lon: 35.896W

Elev: 502

StdP: 95.44

Time Zone: -3.00 (W03)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	18.1	13.8	10.4	29.0	14.2	10.8	28.8	8.8	25.3	8.2	25.0	1.8	130	0.354

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	9.2	32.9	21.4	32.0	21.5	31.2	21.5	23.9	27.8	23.5	27.5	23.1	27.1	4.8	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	19.0	24.9	22.6	18.4	24.3	22.2	17.9	23.9	74.2	27.9	72.5	27.4	71.3	26.8	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
8.4	7.9	7.3	DB	16.1	34.2	0.8	1.0	15.5	34.9	15.1	35.5	14.7	36.0	14.1	36.7	
			WB	15.7	25.9	0.7	1.4	15.3	26.9	14.9	27.7	14.5	28.5	14.0	29.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.8	25.8	25.9	26.2	25.9	24.8	23.5	22.6	22.8	23.9	25.0	25.9	26.1
	DBStd	1.65	1.00	1.03	1.05	1.25	1.07	1.10	1.02	1.02	1.09	0.99	0.83	1.02
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5417	491	446	501	477	458	404	389	398	416	464	476	498
	CDD18.3	2376	233	213	242	227	199	154	131	140	166	206	226	239
	CDH23.3	14703	1618	1483	1694	1523	1108	647	447	626	902	1299	1630	1725
	CDH26.7	4906	607	557	646	558	298	108	45	95	224	436	648	684
Wind	WSAvg	4.1	4.4	4.1	3.9	3.5	3.4	3.4	3.5	3.9	4.4	4.7	4.9	4.7
Precipitation	PrecAvg	863	60	71	100	94	103	142	111	67	33	13	11	24
	PrecMax	1503	401	247	269	229	347	301	278	188	148	53	52	63
	PrecMin	443	5	9	4	7	10	26	0	0	6	1	0	0
	PrecStd	274	75	65	57	57	68	69	59	42	35	11	13	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.9	33.1	33.2	34.1	31.2	30.0	28.9	29.5	31.0	32.0	33.0	33.1
		MCWB	21.3	21.3	21.6	21.6	21.7	21.3	20.8	20.3	21.0	21.3	20.9	21.4
	2%	DB	31.9	31.9	32.1	32.8	30.2	28.9	27.9	28.8	29.9	31.0	31.9	32.0
		MCWB	21.6	21.8	21.9	21.6	21.9	21.3	20.7	20.2	20.7	20.9	21.2	21.4
	5%	DB	31.0	31.0	31.1	31.1	29.6	28.0	27.0	27.9	29.0	30.1	31.0	31.1
		MCWB	21.4	21.7	22.1	21.6	21.9	21.1	20.4	20.2	20.5	21.0	21.1	21.3
	10%	DB	30.0	30.1	30.1	30.0	28.2	27.0	26.0	26.9	28.1	29.1	30.2	30.2
		MCWB	21.5	21.9	22.2	22.1	21.9	21.1	20.6	20.2	20.4	20.9	21.1	21.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.6	24.2	24.4	24.5	24.1	23.5	22.7	22.2	22.7	23.5	23.1	23.6
		MCDB	28.3	28.1	28.5	27.8	27.1	26.3	25.4	25.5	27.4	28.6	29.1	28.8
	2%	WB	23.0	23.5	23.6	23.7	23.4	22.7	22.1	21.6	22.0	22.2	22.2	22.7
		MCDB	27.5	27.5	28.0	27.4	26.6	25.9	25.0	25.2	26.3	27.5	28.0	27.7
	5%	WB	22.6	23.0	23.2	23.2	23.0	22.3	21.6	21.2	21.5	21.7	21.8	22.3
		MCDB	26.9	27.2	27.5	26.8	26.5	25.3	24.4	25.1	25.7	26.8	27.3	27.0
	10%	WB	22.3	22.6	22.8	22.9	22.6	21.9	21.2	20.8	21.2	21.4	21.6	22.0
		MCDB	26.3	26.7	27.0	26.4	25.9	24.7	23.9	24.7	25.3	26.3	26.8	26.4
Mean Daily Temperature Range	5% DB	MDBR	9.4	9.2	9.2	9.0	8.1	7.6	7.3	8.6	9.2	9.9	10.3	9.6
		MCDBR	10.5	10.4	10.5	11.3	9.6	9.1	9.1	10.0	10.3	10.7	10.9	10.6
		MCWBR	2.5	2.3	2.4	2.5	2.9	3.1	3.3	3.1	2.8	2.8	2.2	2.2
	5% WB	MCDBR	9.1	8.8	8.9	8.4	7.8	7.3	7.0	8.1	8.8	9.7	10.1	9.5
		MCWBR	2.8	2.8	2.6	2.8	3.0	3.2	3.4	3.5	3.1	3.1	2.5	2.5
Clear-Sky Solar Irradiance	taub	0.373	0.378	0.382	0.378	0.360	0.360	0.361	0.360	0.379	0.385	0.375	0.387	
	taud	2.494	2.486	2.478	2.497	2.549	2.530	2.509	2.472	2.395	2.401	2.441	2.428	
	Ebn at Noon	966	957	939	916	905	889	895	919	927	938	956	949	
	Edh at Noon	115	116	115	108	98	97	101	109	123	125	120	122	
All-Sky Solar Radiation	RadAvg	5.95	5.91	5.86	5.41	4.94	4.46	4.69	5.29	5.84	6.21	6.43	6.14	
	RadStd	0.40	0.31	0.27	0.30	0.30	0.31	0.32	0.28	0.30	0.35	0.35	0.24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	-0.47	-0.30	-0.65	-0.95	-0.46	-0.33	N/A	N/A	-218	-218

Nomenclature: See separate page